inference_mode#

https://pytorch.org/docs/stable/generated/torch.autograd.grad_mode.inference

Description:

Context manager that enables or disables inference mode. InferenceMode is analogous to no_grad and should be used when you are certain your operations will not interact with autograd (e.g., during data loading or model evaluation). Compared to no_grad, it removes additional overhead by disabling view tracking and version counter bumps. It is also more restrictive, in that tensors created in this mode cannot be used in computations recorded by autograd. This context manager is thread-local; it does not affect computation in other threads. Also functions as a decorator.

Function Signature

```
class torch.autograd.grad_mode.inference_mode(mode=True)
[source]#

Parameters

Example::

clone()[source]#

Return type
```

Returns:

inference_mode

Code Examples

```
>>> import torch
>>> x = torch.ones(1, 2, 3, requires_grad=True)
>>> with torch.inference_mode():
...     y = x * x
>>> y.requires_grad
False
>>> y._version
Traceback (most recent call last):
File "<stdin>", line 1, in <module>
RuntimeError: Inference tensors do not track version counter.
>>> @torch.inference_mode()
... def func(x):
...     return x * x
>>> out = func(x)
>>> out.requires_grad
False
>>> @torch.inference_mode()
... def doubler(x):
...     return x * 2
>>> out = doubler(x)
>>> out.requires_grad
False
```

Notes & Warnings

NOTE

Note Inference mode is one of several mechanisms that can locally enable or disable gradients. See Locally disabling gradient computation for a comparison. If avoiding the use of tensors created in inference mode in autograd-tracked regions is difficult, consider benchmarking your code with and without inference mode to weigh the performance benefits against the trade-offs. You can always use no_grad instead.

NOTE

Note Unlike some other mechanisms that locally enable or disable grad, entering inference_mode also disables forward-mode AD.

Detailed Documentation

Documentation

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mode (bool or function) – Either a boolean flag to enable or disable inference mode, or a Python function to decorate with inference mode enabled.

Create a copy of this class